Temperature Anomalies: Decomposition and Forecasting

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1 Introduction

This analysis examines long-term movements, seasonal patterns, and variability in monthly Northern Hemisphere surface temperature anomalies from GISTEMP.

We first explore the observed series and compare classical additive and multiplicative decompositions. We then use STL to examine fixed and time-varying seasonal specifications, interpreting the estimated trend, seasonal component, and remainder.

The forecasting analysis compares simple exponential smoothing and Holt's method using observations from January 1950 to December 2010. Seasonal adjustment for this stage is estimated using only observations dated within the training period.

Forecasts from the end of 2010 are evaluated against subsequent observations over 2011–2025. We report prediction intervals and cumulative mean absolute errors over the first 1, 5, 10, and 15 years after the forecast origin.

The analysis describes statistical patterns and the performance of extrapolative forecasting methods; it does not identify the physical causes of temperature changes.

2 Data Preparation

The dataset contains monthly Northern Hemisphere land–ocean temperature anomalies relative to the 1951–1980 reference period, expressed in degrees Celsius.

Each row represents a calendar year. We extract the twelve monthly columns, excluding annual and seasonal summaries, and arrange the observations chronologically in a monthly time series.

The sample covers January 1880 to December 2025, with 1,752 monthly observations and no missing monthly values. Negative anomalies indicate temperatures below the reference-period average for the corresponding calendar month; positive anomalies indicate temperatures above it.

3 Temperature Dynamics

We examine the monthly anomalies to identify long-term movements and short-term fluctuations.

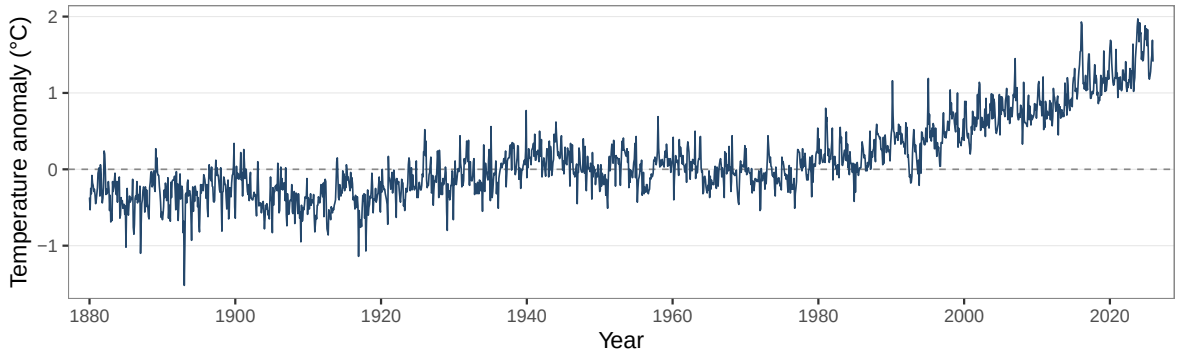


Figure 1. *Monthly Northern Hemisphere land–ocean temperature anomalies, 1880–2025. The dashed line marks the 1951–1980 reference level.*

The series shows a long-term increase interrupted by substantial monthly fluctuations. Warming is uneven: a relatively flat period around the middle of the twentieth century is followed by a more pronounced rise from approximately the 1970s onward. Recent observations remain above the reference level.

4 Persistence and Variability

We examine autocorrelation up to 48 months and the standard deviation within centered 120-month windows.

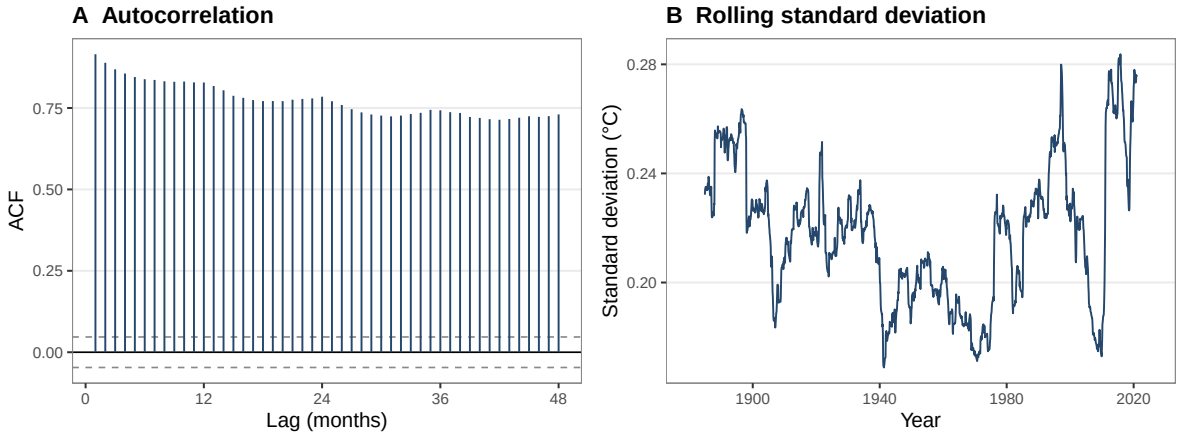


Figure 2. *Autocorrelation of monthly anomalies (A) and rolling standard deviation over centered 120-month windows (B). Dashed ACF bands are approximate white-noise reference bounds.*

Autocorrelations remain high across the displayed lags, indicating strong persistence consistent with the long-term movement. This is not a formal test of non-stationarity.

The rolling standard deviation varies over time without a steady increase across the full sample. Since each window also contains local trend movements, this measure does not isolate the variance of the irregular component.

5 Classical Decomposition

We compare additive decomposition, $Y_t = T_t + S_t + R_t$, with multiplicative decomposition, $Y_t = T_t S_t R_t$. Both estimate the trend using centered moving averages and impose a repeating seasonal pattern.

Temperature anomalies include negative values and cross zero, making proportional seasonal factors difficult to interpret. The multiplicative specification is retained as a diagnostic comparison.

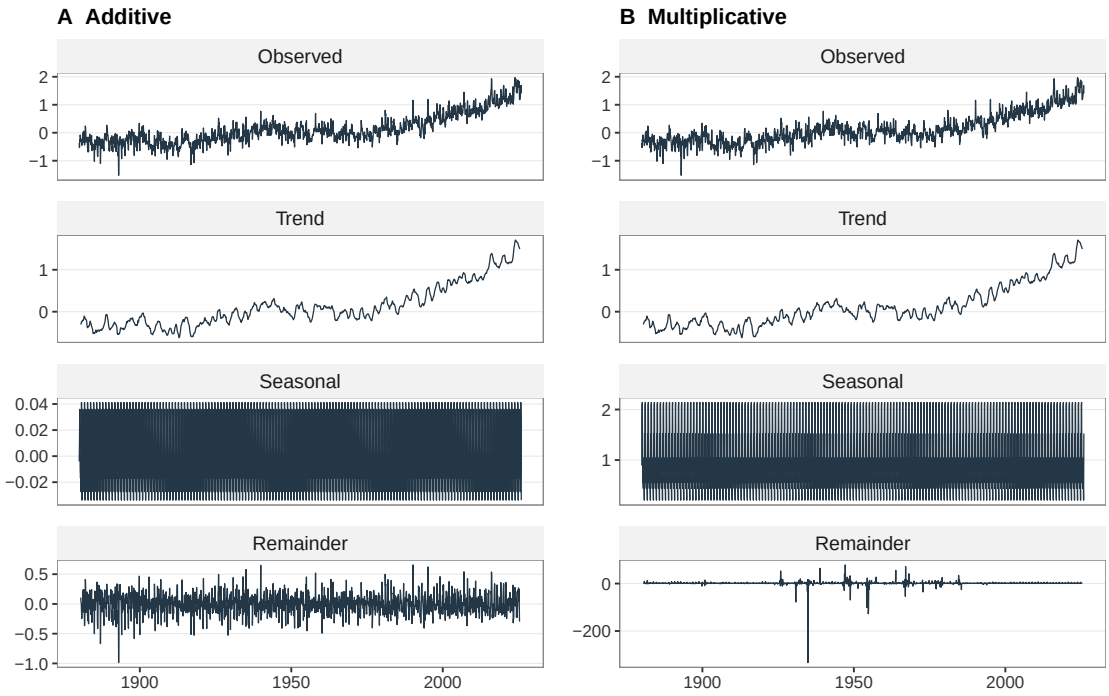


Figure 3. *Classical additive (A) and multiplicative (B) decompositions. Vertical scales vary across panels. Additive components are in degrees Celsius; multiplicative seasonal and remainder components are ratios.*

The trend estimates coincide because both specifications use the same moving-average filter. They show an uneven long-term increase, with stronger warming in the later decades.

The multiplicative remainder exhibits extreme spikes when its denominator, the product of trend and seasonal components, approaches zero. These ratios are sensitive to the chosen anomaly baseline. We therefore use an additive representation for subsequent analysis. The additive remainder still requires examination; its appearance alone does not establish white noise.

6 Fixed and Time-Varying Seasonality

We compare robust STL with a fixed annual seasonal pattern (`s.window = "periodic"`) and a pattern allowed to evolve over time (`s.window = 13`). Robust fitting reduces the influence of unusually large remainder values.

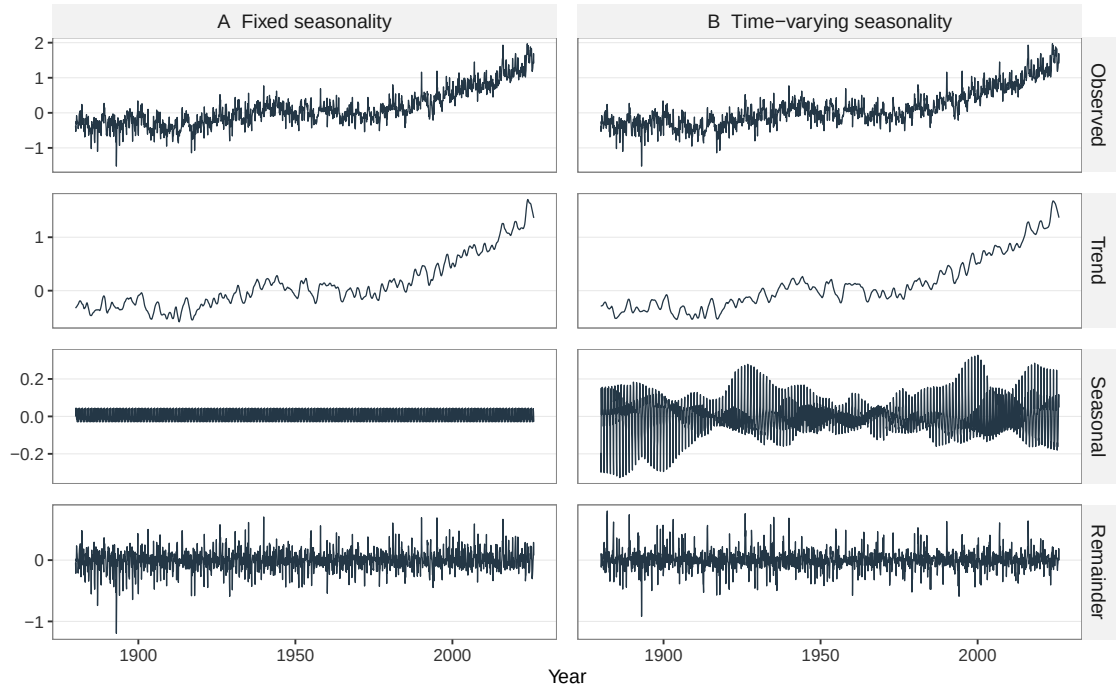


Figure 4. *Robust STL with fixed (A) and time-varying (B) seasonality. Components are expressed in degrees Celsius. Each row shares a vertical scale across the two specifications.*

Both specifications recover a similar long-term warming pattern. The flexible specification allocates more variation to the seasonal component, but this alone does not establish that the variation represents a stable seasonal signal.

We retain fixed seasonality as a parsimonious working specification for forecasting. This is a modelling choice, rather than evidence that seasonality is constant or that the flexible fit overfits.

7 Seasonal Pattern and Remainder

We examine the monthly seasonal effects and the remainder from classical additive decomposition.

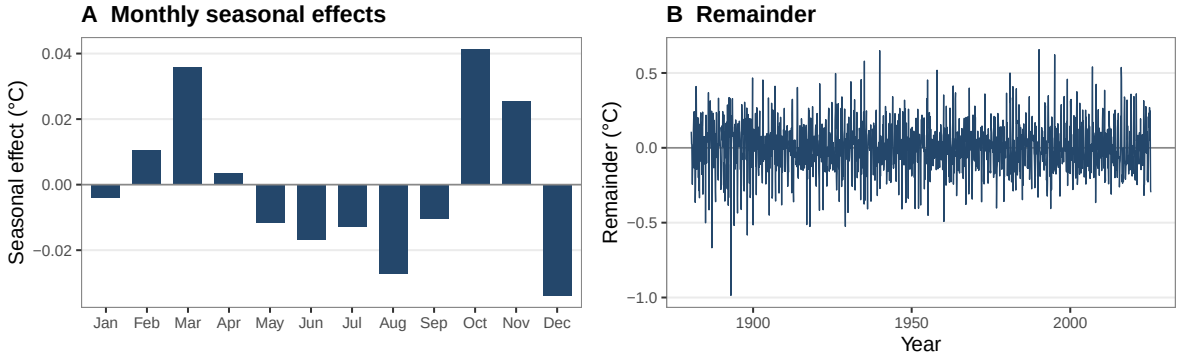


Figure 5. *Monthly seasonal effects (A) and remainder over time (B) from classical additive decomposition, in degrees Celsius.*

The estimated monthly effects are small, with positive peaks in March and October and negative effects in several summer and winter months. These are patterns in temperature anomalies, not the annual cycle of absolute temperatures; their physical causes cannot be inferred from this decomposition.

The remainder fluctuates around zero with occasional large deviations, particularly early in the record. These represent variation left unexplained by the estimated trend and fixed seasonal pattern.

8 Exponential Smoothing

We seasonally adjust the January 1950–December 2010 sample using robust periodic STL fitted to that sample only. We then compare simple exponential smoothing (SES), which updates a level, with Holt's method, which also updates a trend.

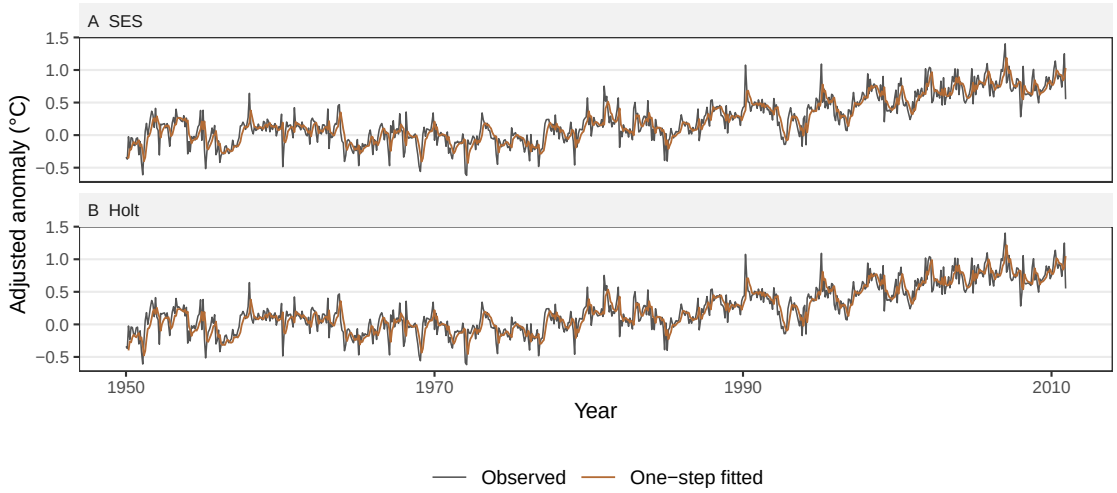


Figure 6. *Seasonally adjusted training observations and one-step-ahead fitted values from SES and Holt, 1950–2010. Both panels share the same axes.*

The estimated level smoothing parameters are $\alpha = 0.419$ for SES and $\alpha = 0.449$ for Holt. Holt's trend smoothing parameter is $\beta = 0.011$. Parameters are selected by minimizing the in-sample sum of squared one-step-ahead errors.

These fitted values use preceding observations, but their smoothing parameters and seasonal adjustment are estimated from the full training sample. Their fit therefore does not establish performance on unseen data.

We next fix Holt's level smoothing parameter at 0.1 and 0.9, re-estimating the trend smoothing parameter in each case.

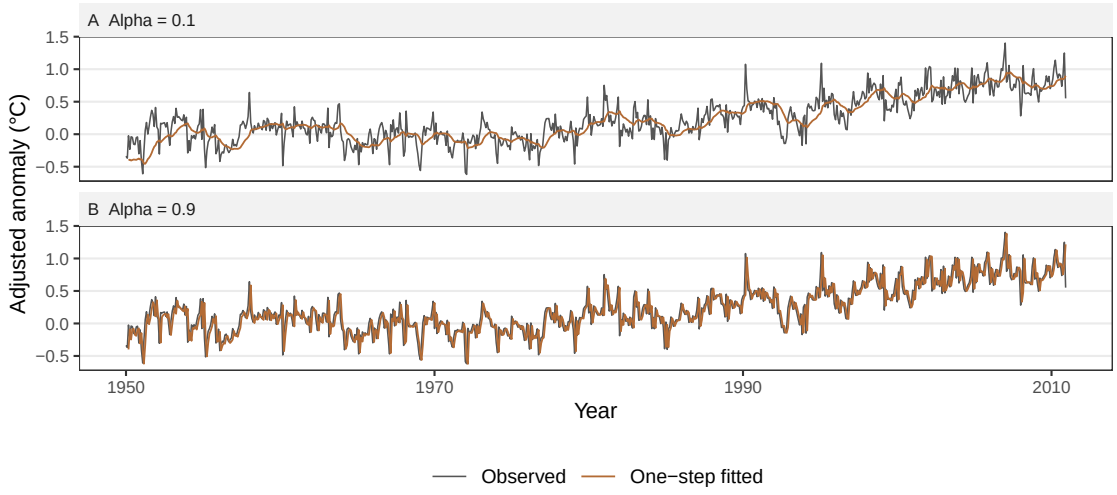


Figure 7. *Holt one-step-ahead fitted values with alpha fixed at 0.1 and 0.9. Beta is estimated separately for each specification; both panels share the same axes.*

A larger alpha gives the latest observation greater weight in the level update. The comparison also reflects the separately estimated beta values: 0.030 for alpha = 0.1 and 0.005 for alpha = 0.9. Responsiveness alone does not establish better forecast accuracy.

9 Forecast Evaluation

The Holt model estimated on the seasonally adjusted 1950–2010 sample produces forecasts for January 2011 through December 2025, without re-estimation. Test observations are adjusted using the monthly seasonal effects estimated on the training sample.

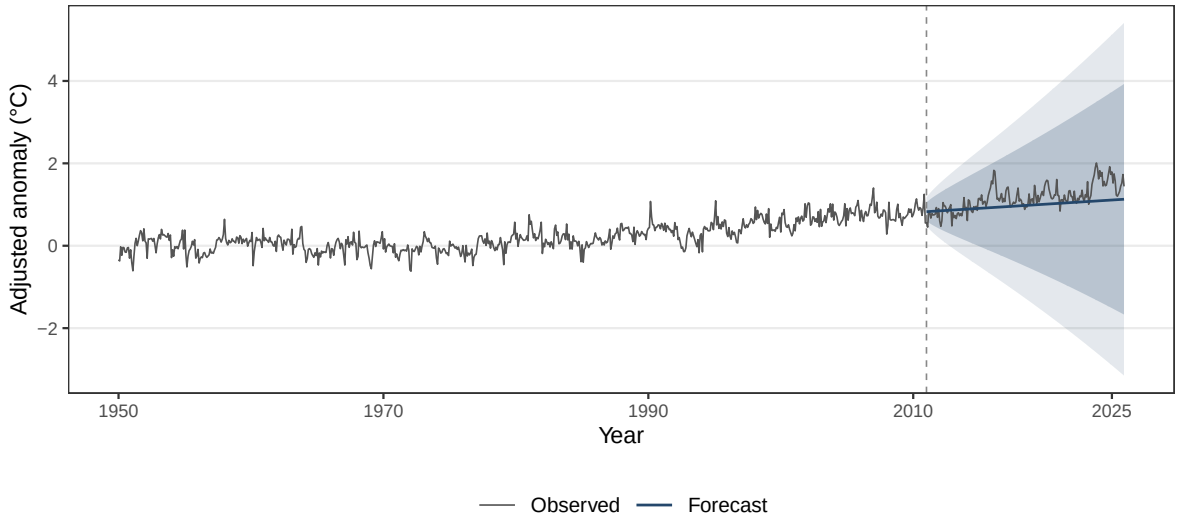


Figure 8. *Holt forecasts from the end of 2010, with 80% and 95% prediction intervals. Observations are seasonally adjusted using training-sample estimates. The dashed line marks the start of the test period. Intervals are conditional on the fitted model and exclude uncertainty in seasonal adjustment.*

The point forecasts extend the estimated trend, while observations in the later test years generally lie above the forecast path. Prediction intervals widen substantially over the fifteen-year horizon.

Forecast accuracy is summarized using mean absolute error (MAE) over the first 12, 60, 120, and 180 test months. Each value averages all monthly absolute errors up to the indicated horizon; it is not the error at the endpoint alone.

Table 1: Cumulative forecast MAE from January 2011.

Evaluation horizon	Months	MAE (°C)
1 year	12	0.120
5 years	60	0.157
10 years	120	0.200
15 years	180	0.239

Cumulative MAE increases from 0.120 °C over the first year to 0.239 °C over fifteen years. Average absolute errors are therefore higher when later test observations are included. These results describe a single forecast origin and do not establish how accuracy would vary across different estimation periods.

10 Discussion and Limitations

Additive decomposition is appropriate for temperature anomalies, which include negative and near-zero values. STL illustrates how allowing seasonal effects to vary changes the allocation between seasonality and remainder. The Holt forecasts extend the estimated trend but generally underpredict anomalies in the later test years.

The evaluation uses one chronological split and a dataset retrieved after the forecast origin, rather than the data vintage available in 2010. Prediction intervals depend on the model assumptions and exclude uncertainty in seasonal adjustment. The analysis provides a statistical description and forecasting illustration, without identifying the causes of temperature changes.